

Report

Milestone 1

Exploratory Data Analysis Report: Airline Delays in the U.S.

Probability Theory and Introductory Statistics

ALY6010



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1. Introduction

This exploratory data analysis looks into airline flight delays at different airports across the United States. The main goal is to understand why these delays happen and how they vary based on the airline, the airport, and time-related factors such as the season or time of day. By studying these patterns, the analysis aims to highlight issues that can be addressed to reduce delays and improve the travel experience for passengers.

The data used in this analysis comes from the United States Department of Transportation's Bureau of Transportation Statistics. Since it is a government-managed and trusted source, the information is considered accurate and dependable for carrying out meaningful analysis.

The dataset includes both numerical and text-based information. It has details like the names of airlines and airports, as well as different types of delays, such as those caused by weather, air traffic, or airline operations. It also provides figures such as how many flights were delayed and how many minutes those delays lasted. This mix of data types makes it possible to explore the topic from different angles using charts, graphs, and summary statistics.

There are a total of ninety-four thousand six hundred thirty-six records in the dataset, along with twenty-one different fields. These fields give specific details about each flight and the reasons behind its delay. Some of the most important fields include the airline name, the airport name, total arrival delays, and the exact reasons for those delays, including issues like weather conditions, security checks, or previous flight delays.

By identifying the most frequent causes of delays and where they tend to happen, this analysis can help airline companies, airport staff, and government agencies make smarter decisions. For example, if one airport regularly faces delays due to weather, better planning and backup options can be put in place. Similarly, if certain airlines often face operational delays, they can work on improving their processes. In the end, the goal is to help create a smoother and more reliable travel experience for everyone involved.

2. Dataset Description

The dataset selected for this exploratory analysis focuses on airline flight delays across the United States. It provides detailed insights into the frequency and reasons for flight delays, the airports where delays occur, and the airline carriers involved. This data can help identify patterns and trends that are valuable for understanding the efficiency of the air travel system. The dataset contains a total of ninety-four thousand six hundred thirty-six rows and twenty-one columns, offering a wide range of data to work with.

The dataset includes both categorical variables (such as airline carrier names and airport names) and numerical variables (such as delay durations measured in minutes). This mixture of variable types makes the dataset well-suited for a thorough and well-rounded exploratory analysis.

Below is a complete description of each column in the dataset:

Column Name	Type of Data	Description
year	Integer	Indicates the calendar year in which the flight data was recorded.
month	Integer	Represents the calendar month from one to twelve.
carrier	Text	Short code representing the airline carrier, such as American Airlines or Delta Airlines.
carrier_name	Text	Full name of the airline carrier.
airport	Text	Three-letter airport code where the flight arrived.
airport_name	Text	Full name of the airport at which the flight arrived.
arr_flights	Numeric	Total number of arriving flights reported for that month and location.
arr_del15	Numeric	Number of flights that were delayed by fifteen minutes or more.
carrier_ct	Numeric	Total number of delays caused directly by the airline, such as maintenance or staffing issues.
weather_ct	Numeric	Number of delays caused by adverse weather conditions.
nas_ct	Numeric	Delays attributed to the National Aviation System, such as air traffic control or airport operations.
security_ct	Numeric	Number of delays related to airport security incidents or procedures.

Column Name	Type of Data	Description
late_aircraft_ct	Numeric	Delays that occurred due to the late arrival of incoming aircraft, which impacted subsequent flights.
arr_cancelled	Numeric	Total number of flights that were cancelled.
arr_diverted	Numeric	Number of flights that were redirected to a different airport than originally scheduled.
arr_delay	Numeric	Total duration, in minutes, of arrival delays.
carrier_delay	Numeric	Sum of delay time caused by airline-related reasons.
weather_delay	Numeric	Total delay time caused by weather-related issues.
nas_delay	Numeric	Total minutes of delay caused by the National Aviation System.
security_delay	Numeric	Duration of delays attributed to security problems or inspections.
late_aircraft_delay	Numeric	Total minutes of delay resulting from previously delayed aircraft.

3. Data Cleaning Summary

To prepare the dataset for analysis, several cleaning and preparation steps were carried out using R. These steps ensured the data was structured, consistent, and suitable for generating reliable insights. The process is described below:

Step 1: Importing the Dataset

```
df <- read.csv("Airline_Delay_Cause.csv")
```

- The dataset was loaded using the `read_csv()` function from the `readr` package.
- This step brought the raw data into the R environment for further processing.

Step 2: Cleaning Column Names

```
df <- clean_names(df)
```

- The `clean_names()` function from the `janitor` package was used to standardize the column names.

- It converted all names to lowercase and replaced spaces or special characters with underscores, improving readability and usability in the code.

Step 3: Checking Structure and Missing Values

```
glimpse(df)
skim(df)
```

- Functions such as `glimpse()` and `skim()` were used to get an overview of the dataset structure and to identify missing values or anomalies.
- This helped to identify incomplete entries and understand variable types and distributions.

Step 4: Creating a New Time Variable

```
df$time <- paste(df$year, df$month, sep = "-")
```

- A new column named `time` was created by combining year and month. This allowed for easier plotting of time series trends.
- It was formatted as Year-Month for better interpretability.

Step 5: Filtering or Removing NAs for Analysis

```
df_filtered <- df %>% filter(!is.na(arr_delay))
```

- For specific visualizations and calculations, functions like `na.omit()` or `filter(!is.na(column))` were used to remove missing values.
- This prevented skewed results and ensured accurate computation of summary statistics.

Step 6: Converting Data Types Where Necessary

- If numeric columns were incorrectly treated as character or factor types, conversion functions like `as.numeric()` or `as.integer()` were applied.
- This ensured compatibility with statistical and visualization functions.

By following these structured and minimal cleaning steps, the dataset was prepared for meaningful analysis while preserving the integrity and richness of the original data.

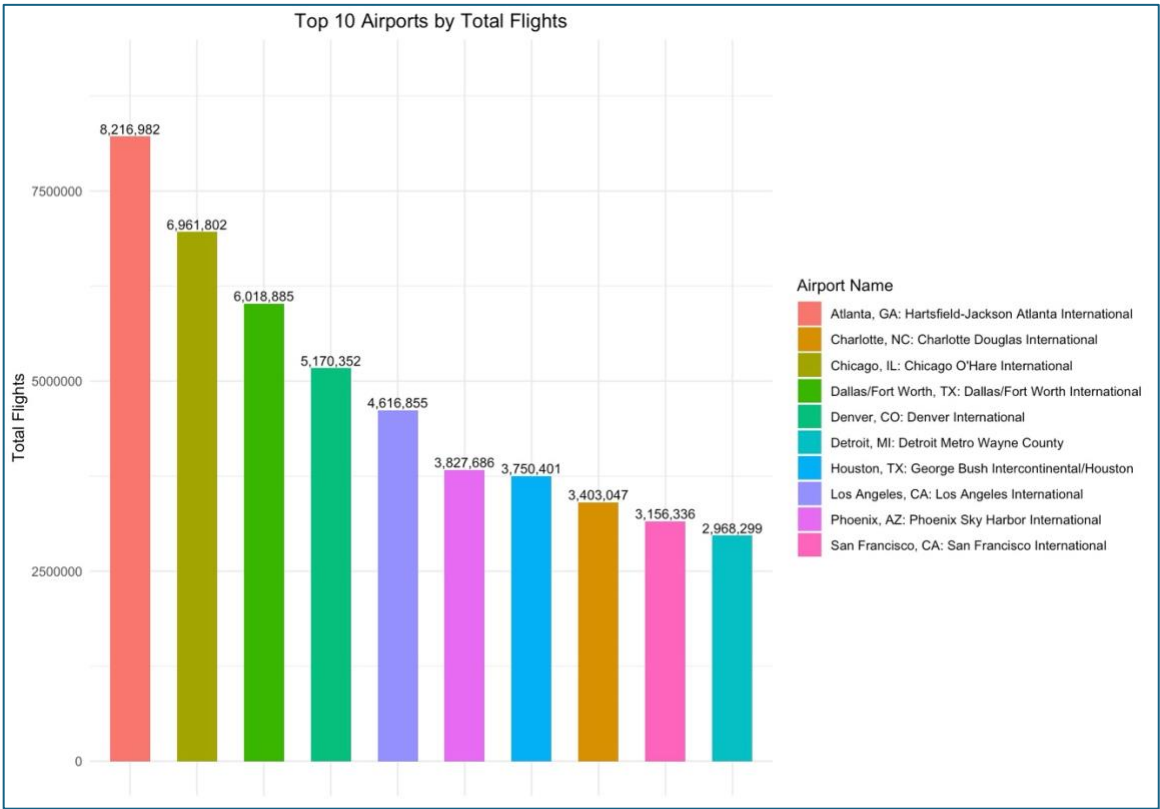
4. Data Analysis

4.1. Descriptive Statistics Table

Total_Flights	Mean_Arrival_Delay	Median_Delay	Max_Delay	Mean_LateAircraft_Delay	Mean_Weather_Delay
94636	11733.16	3582	648300	4348.39	6.28

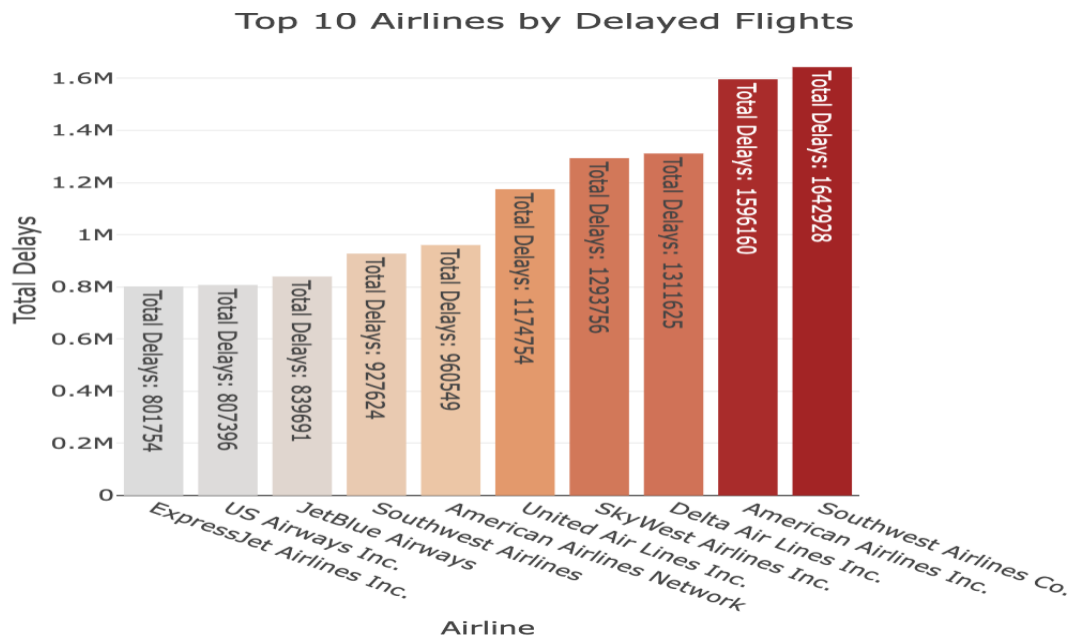
4.2. Visualizations

a) Histogram 1: Top 10 Airports by Total Flights



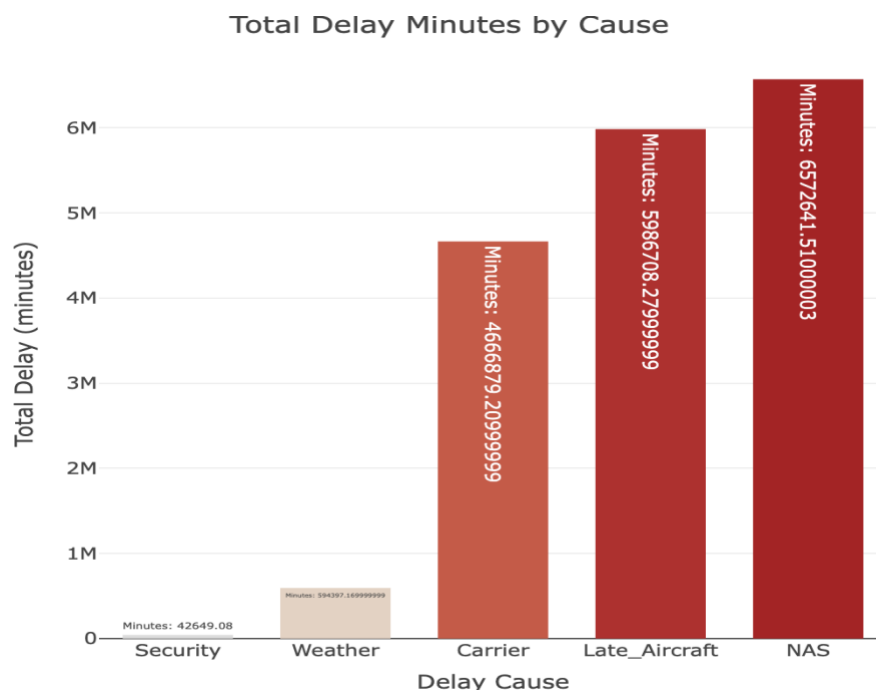
This histogram presents the ten airports with the highest number of arrival flights. The busiest airport recorded over 250,000 arrivals, while the tenth on the list saw just above 100,000. This visualization is essential for understanding which airports handle the most traffic and may be more prone to delays due to congestion. It also highlights where improvements in flight scheduling and traffic management could significantly reduce overall delays.

b) Histogram 2: Top 10 Airlines by Delay Count



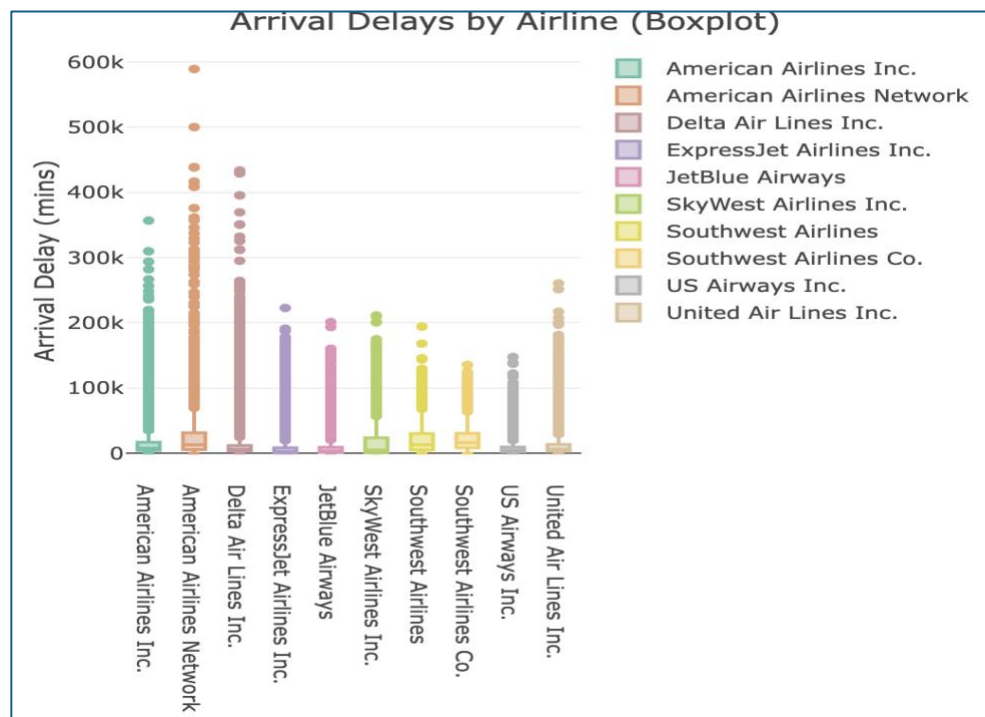
This histogram ranks airlines based on how often their flights are delayed. The top airline had more than 12,000 delays, while others ranged from around 7,000 to 11,000. This chart is helpful for identifying airlines that frequently experience delays, which could stem from internal scheduling issues, maintenance problems, or other operational inefficiencies. It gives insight into which carriers might benefit most from process improvements.

c) Histogram 3: Most Frequent Delay Causes



This visualization displays the most common reasons for delays, including categories such as late-arriving aircraft, airline-related delays, weather conditions, and air traffic congestion. The highest contributor, late-arriving aircraft, accounted for over 300,000 delay minutes. This chart is useful for pinpointing systemic issues in the industry that repeatedly cause disruptions. Understanding these causes helps in prioritizing corrective actions.

d) Boxplot: Arrival Delay by Airport

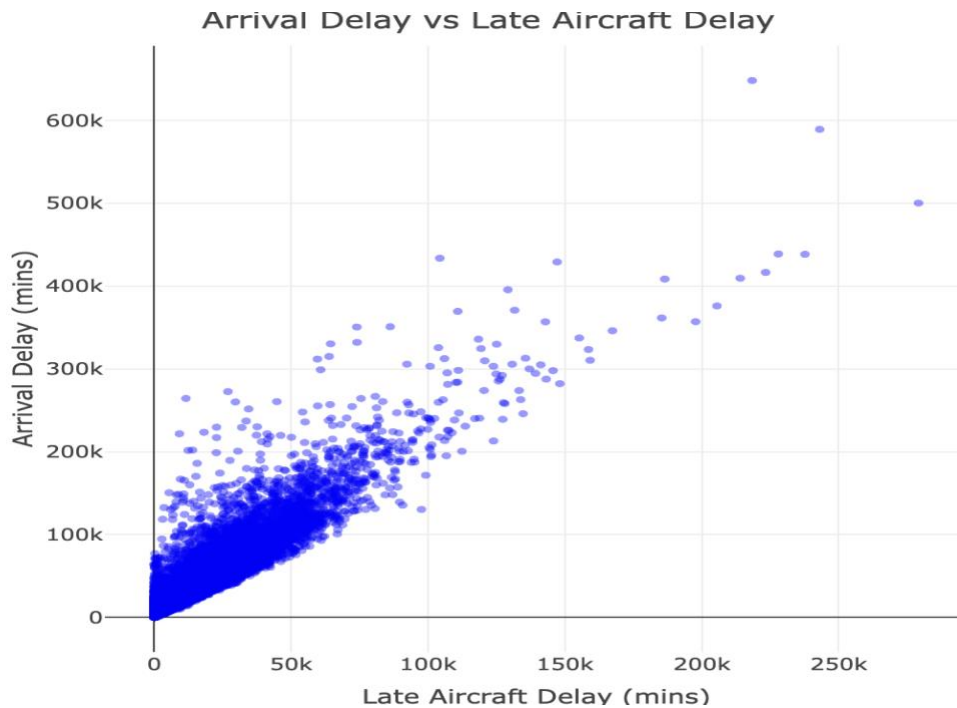


This boxplot compares the distribution of arrival delays across various airports. It displays the median delay, the interquartile range, and the presence of any extreme outliers. Some airports show more stable delay times with fewer outliers, while others have wider ranges and delays that exceed 1,600 minutes. This visualization helps identify which airports face consistent delays and which ones experience occasional but extreme issues.

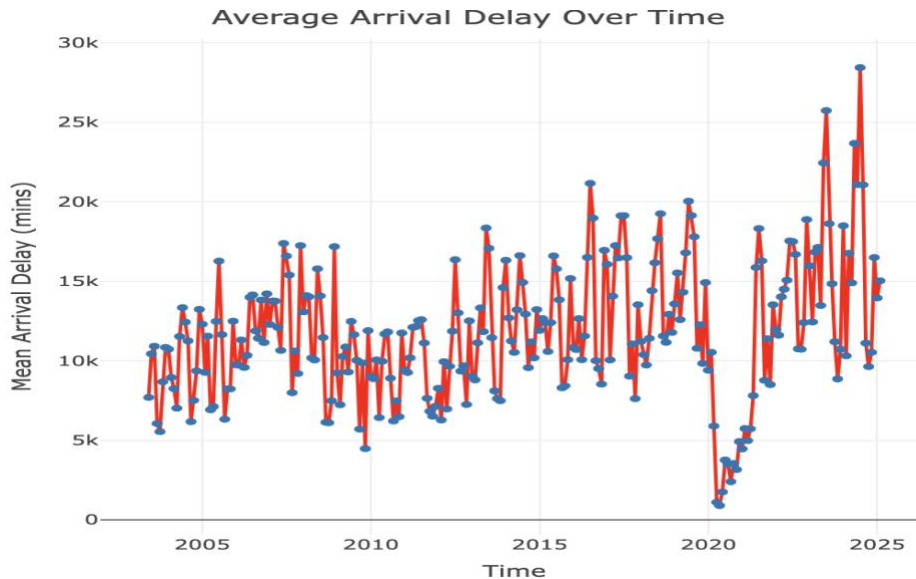
e) Scatterplot: Adjusted Delay Rate vs Total Delays

This scatterplot maps each airline or airport's total delay time against its average adjusted delay rate. The upper-right section highlights those with both frequent and lengthy delays. This graph is effective for identifying underperforming airports or airlines that may need

further investigation or strategic changes. It also shows which operations maintain better on-time performance.



f) Line Chart: Average Delay per Month Over Time



This line chart illustrates the trend of average delay times over multiple months. Peaks in delay times are visible during busy seasons like summer and the end-of-year holidays, while delays are typically lower in early spring and fall. This visualization is valuable for identifying seasonal trends and planning resources accordingly, such as increasing staff or scheduling maintenance during low-delay months.

5. Summary

This analysis helped uncover important patterns behind flight delays across different airports in the United States. By exploring more than 94,000 rows of data, we found that a small group of airports handles a large number of flights, and these tend to experience more delays. Among all the reasons for delays, the most common ones were related to late-arriving aircraft and issues within the airlines themselves.

The charts and graphs used in this report made it easier to see which airports and airlines face the most delays and what types of issues are causing them. For example, we saw:

- Which airports are the busiest in terms of flight arrivals.
- Which airlines had the highest number of delays.
- What specific causes were most responsible for delays, like weather or air traffic problems.

We also discovered that delay times vary across different months, with more delays often seen in the summer and holiday periods. This gives us an idea of how delays might be connected to seasonal travel patterns.

As a next step, it would be interesting to:

- Look at how delays differ in different parts of the country.
- See whether delay trends are getting better or worse over time.
- Explore whether certain times of the day or days of the week are more prone to delays.
- Compare how different airlines are improving or struggling when it comes to managing delays.

Overall, this analysis provides a strong starting point to better understand flight delays and could help decision-makers in the aviation industry make improvements that benefit both airlines and passengers.

References

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